

PLAYING TOUGHER FOR ONE'S SIZE: POSITIONAL PHYSICALITY IN THE NHL

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Introduction

Physical presence takes shape in a hit, a puck battle, or a trip into traffic around the net. How much of that presence exceeds what a player's size would suggest? We develop calibrated size above expected (CSAx) to measure playing tougher for one's size, giving scouts and teams a common language for comparing physical profiles.

Methods

We combine NHL events with All Three Zones' manually tracked puck plays from 2021-22 through 2024-25, covering 1,485 forward-seasons and 829 defenseman-seasons. Each player's pattern of direct contact and indirect physicality suggests an expected body size, estimated through separate positional ridge regressions. We calibrate held-out predictions against each player's height and weight, then standardize the differences within position and season. Higher CSAx indicates more physical presence than expected for that frame. We examine earlier scouting descriptions, special-teams assignments, playoff contact, and subsequent NHL participation.

Results

Independent scouting descriptions align with CSAx: players described as actively physical receive higher scores in both positions (Figure 1).

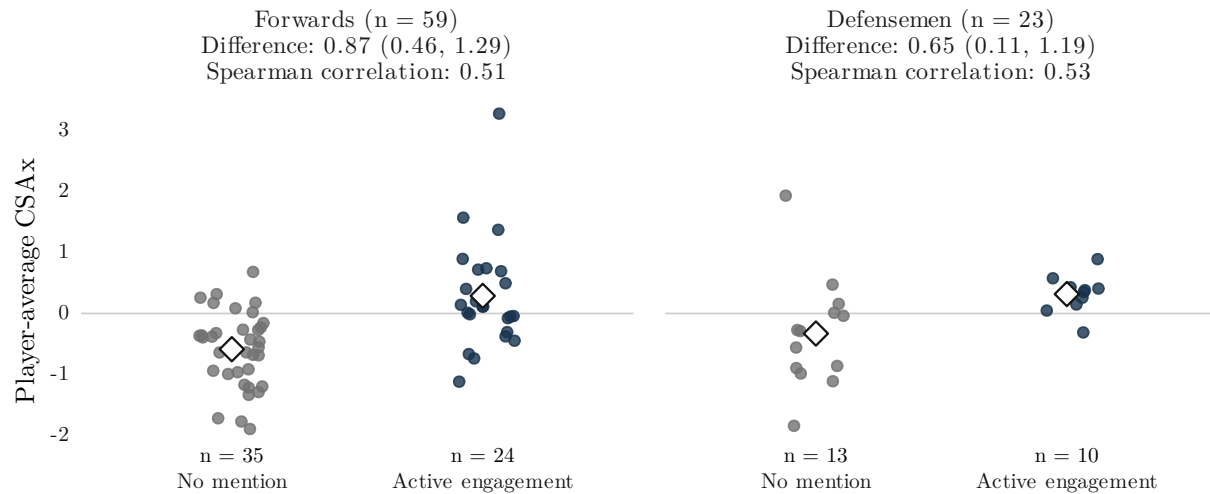


Figure 1. Player-average CSAx by scouting description. Points represent players; diamonds mark means. Differences show 95% intervals. No mention does not establish soft play.

These profiles also correspond to distinct roles: higher-scoring players spend less of their ice time on the power play and more killing penalties. Among forwards, higher CSAx accompanies a smaller proportional playoff increase in delivered hits. This comparison uses regular-season games excluded from score construction and each team’s first four playoff games; the full-postseason check agrees in direction. Finally, higher CSAx accompanies a greater chance of reaching 300 NHL minutes next season, with no clear increase in minutes per game among returners (Table 1).

Table 1. Adjusted CSAx +1 versus -1 comparisons (two standard deviations apart). Parentheses show 95% intervals; pp denotes percentage points. Ice time covers returners.

Outcome	Forwards	Defensemen
Power-play share (pp)	-1.22 (-1.78, -0.67)	-0.88 (-1.51, -0.25)
Penalty-kill share (pp)	1.40 (0.74, 2.06)	1.31 (0.74, 1.89)
Playoff hit-rate change (ratio)	0.78 (0.69, 0.88)	0.90 (0.77, 1.05)
Next-season continuation (pp)	2.92 (0.03, 5.81)	4.23 (0.60, 7.86)
Next-season ice time (min/game)	-0.18 (-0.36, 0.00)	-0.14 (-0.43, 0.14)

The hit-rate ratio compares playoff-to-baseline rate changes; values below one indicate a smaller proportional increase.

Conclusion

CSAx offers teams a frame-relative view of physical style, while the roster results distinguish continued participation from playing time among returners. Selective coverage, one scouting rater, weak 2024-25 defensive size prediction, and uncertain defensive playoff differences limit generalization. Associations are observational; intervals condition on estimated scores.